Section A : Data Fundamentals

Database

A database is an organised collection of data stored so it can be easily accessed managed and updated

it helps store large amount of information safely and quickly.

Example:

A school storing student records
A bank storing customer account details

Data Warehouse

a datawarehouse is a system used to store large amounts of historical data for analysis

Difference.

DATA LAKE

* a data lake stores raw data (Structured, and unstructured) in its original format.

Difference :

Data Warehouse = Structured & Organized
Data Lake = raw & flexiable

Example :
Storing videos, emails, and social media data

Data Mart

A smaller part of a data warehouse focused on one department

Example :
Sales team using only sales data.

Type of Data

Structured . Organised (tables) e.g. Excel sheet.
Semi-Structured , Some Structured . e.g Json file
Unstructured . No Structure e.g. Videos, images

Data Modelling

. Designing how data is stored and related
importance . helps Organized data properly

Types : * Conceptual model
* Logical Model

11. Artificial intelligence

* Machines that act like humans.
* Narrow AI → (Special task) (Siri)
* General AI → human level thinking

12. Machine Learning

* Machines learn from data.

type . Supervised-Labeled data
. Unsupervised-no labels
. Reinforcement-Learning by rewards

13. Deep Learning

* advanced ML using neural networks

> Neural Networks
System that mimic the human brain.

14. NLP

* Machines understand human Language.

e.g. chatbots
. voice assistants
. Translation Apps

15. LLM

* AI trained on large text data

e.g. . Chat GP - Answering questions
. Claude - writing & analysis

16. Generative AI

* AI that creates content

e.g. . CHATGPT - Text
. DALL·E - Image
· midjourney — Art

17. Training Dataset

* Data used to train AI.

N.B: Bad Data → wrong results

18. Computer Vision

* machines understand images

e.g. face recognition
· Self driving cars

SECTION C : APIs & Tools

19. API

* Allows apps to communicate

e.g. weather app getting data from server

20. API key

Secret code to access API

Risk . if exposed - Others misuse it

Section E (Short Answers)

35. Interesting Concept
E.g. AI because it can think like humans

36. Difficult Concept
E.g: Machine Learning because it is complex

37. Future impact
E.g: AI will help in jobs and daily life

REFERENCES

- Google (2023). Machine Learning Basics Google
- IBM (2023). What is AI? IBM
- Microsoft. (2023). Data Fundamentals

GDPR Law in Europe protecting data

0. Data Ethics

* Using data responsibly

E.g Selling user data
Biased AI decisions
Tracking without consent

31 Data Governance

* Managing data quality & security

32. Bias in AI

* Unfair results

E.g AI hiring system favoring men

33. Roles

- Data Analyst - analyze data
- Data Scientist - build models
- Data Engineer - Create AI tools

Engineer build systems

3.4 Business intelligence

* Using data for decisions

Tools
- Power BI
- Tableau

Cloud Computing

Using internet Servers

types
- LaaS - infrasture
- PaaS - platform
- SaaS - Software

Example: AWS
- Google Cloud
- Microsoft 365

6. Version Control

* ~~takes~~ tracks codes changes
Git most popular Tool
Commit - Save changes
Branch - Separate version

Jupyter Notebook

* Tools for writing & running code.
Language: Python

8 Python

* Programming language
Why popular: Easy & Powerful

Libraries
- Pandas - data analysis
- Numpy - math
- Matplotlib - Charts
- Scikit - learn - ML

Rest API

Standard Way to use APIs

method

- GET - get data
- Post - Send DATA
- Put - update
- Delete - remove

JSON

Data Formats For APIs

E.g. {"name": "John", "Age": 25}

Prompt Engineering

- Writting good instruction for Ai

example.

Bad "Explain"

Good "Explain machine learning in Simple Terms with Example"

MAP Function

Applies a function to each item

Example (python):

```
numbers = [1,2,3]
results = list(map(lambda x: x*2, numbers))
```

* Data Mining
* Finding patterns in large data sets

example: Bank detecting fraud
Stores predicting customer buying habits

8. DBMS
Software used to manage dabases
e.g.
* MySQL → Websites
* PostgreSQL → Complex System
* Oracle → Large Companies
* mongoDB → Flexible data

9. E.T.L. Process of moving data
* Extract - get data
* Transform - Clean data
* Load - Store data
Example: moving data from excel to database

10. Data Formats
* CSV - Simple tables
* JSON - APIs
* Parquet - big data
* Avro - Fast processing